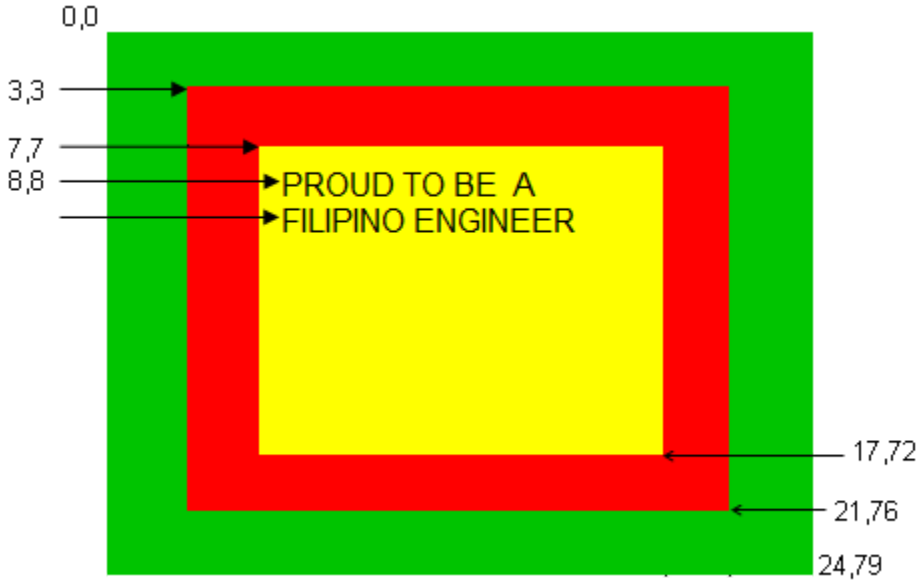
Skills Review

Name: Aaron Ludwig A. Altar **Grade:**

Course/Section: CPE4B **Date: Oct. 7, 2026**

Example No. 11: This program is a combination of Function 09h, Interrupt 10h video function call with scroll down text lines and Cursor Position. It will display the string variables on the coordinates that are specified inside the multiple layers of screen displays. Save the program using sample11.asm as the filename

Setting Coordinates based on the screen layout



Program

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

var1    DB 'PROUD TO BE A ',13,10,'$'
var2    DB 'FILIPINO ENGINEER ',13,10,'$'

.code
start:

        ; 1st layer - green background (0,0 - 24,79)
        mov ah, 07h             ; Function 07h - scroll down window
```

```
        mov bh, 20h             ; background color green
        mov ch, 0               ; start row
        mov cl, 0               ; start column
        mov dh, 24              ; end row
        mov dl, 79              ; end column
        int 10h                 ; BIOS video interrupt - display the window

        ; 2nd layer - red background (3,3 - 21,76)
        mov ah, 07h
        mov bh, 40h             ; background color red
        mov ch, 3
        mov cl, 3
        mov dh, 21
        mov dl, 76
        int 10h

        ; 3rd layer - yellow background (7,7 - 17,72)
        mov ah, 07h
        mov bh, 60h             ; background color yellow
        mov ch, 7
        mov cl, 7
        mov dh, 17
        mov dl, 72
        int 10h

        mov ax, @data           ; initialize the data segment
        mov ds, ax

        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00              ; page 0
        mov dh, 8               ; row 8
        mov dl, 8               ; column 8
        int 10h

        mov ah, 09h             ; Function 09h - display string
        mov dx, offset var1     ; "PROUD TO BE A" at 8,8
        int 21h

        mov ah, 02h
        mov bh, 00
```

```
        mov dh, 9               ; row 9
        mov dl, 8               ; column 8
        int 10h

        mov ah, 09h
        mov dx, offset var2     ; "FILIPINO ENGINEER" at 9,8
        int 21h

        mov ah, 4Ch             ; terminate program
        int 21h

end start
```

Output

